Abstract

Neoclassical microeconomic doctrine treats market price as an unobservable, subjective equilibrium scalar ($P = f(S, D)$), asserting that prices reflect voluntary marginal utility and symmetrical competitive market clearing. This paper presents a complete empirical operationalization of the Universal Price Equation (UPE) from Value Physics, constructing forward biophysical prices and reverse-engineering observed commercial Recommended Retail Prices (RRP) across thirty-four foundational necessities spanning seven essential provisioning sectors in eastern Australia: Food, Agriculture, Poultry, Sugar, Oilseeds, Chemical Inputs, Fertilizers, Liquid Fuels, Power Utilities, Municipal Water/Sanitation, Essential Healthcare, Building Materials, Interior Linings, Rural Reticulation Infrastructure, Spatial Freight Logistics, Agri-Fiber, and Digital Infrastructure. Drawing on official empirical microdata from ABARES, the Australian Bureau of Statistics (ABS), the Australian Energy Market Operator (AEMO), the Australian Competition and Consumer Commission (ACCC), the National Heavy Vehicle Regulator (NHVR), Australian Wool Innovation (AWI), and regional local government authorities, we decompose each price into its physical substrate base (P_b), thermodynamic exergy destruction (P_e), and human metabolic labor difficulty (U, measured in WEST tokens). Solving the inverse UPE for the implied institutional extractive drag fraction (R_a), we establish an objective, non-usurious Recommended Retail Price based on Difficulty (RRP/D). The empirical findings demonstrate that market prices embed an institutional extractive drag (R_a) ranging from 20.8% in owner-operator municipal water cartage up to 79.2% in domestic broadband data transit, establishing that cost-of-living inflation across Australia's core provisioning supply chains is fundamentally an institutional rentier extraction pathology rather than a failure of physical worker productivity.

Keywords: Universal Price Equation , Value Physics , Recommended Retail Price based on Difficulty (RRP/D) , Extractive Drag , Biophysical Economics , Telecommunications Monopoly , Pharmaceutical Pricing , Construction Material Cartels , Agricultural Processing , Infrastructure Pricing , Cement Manufacturing , Agrochemical Cartels , Drywall Duopolies

JEL: D43 , L11 , Q01 , Q40 , R41 , L96 , I11 , Q11 , L74 , L61 , Q12 , L65

Empirical Case Study: Real-World Pricing and RRP/D Fairness Audit — Food, Fuel, Energy, Shelter Materials, Essential Care, Agricultural Inputs, and Digital Infrastructure

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1 Introduction & Epistemological Critique of Subjective Pricing

Standard neoclassical microeconomic doctrine asserts that market price represents the marginal utility of consumers balanced against the marginal cost of production. In this paradigm, prices serve as costless information vectors guiding optimal resource allocation. However, this framework suffers from three fatal physical and institutional blind spots:

1. **Dimensional Incommensurability:** Neoclassical production functions treat capital, energy, and labor as fungible value aggregates denominated in nominal currency, ignoring the thermodynamic reality that physical goods require irreversible exergy destruction (Georgescu-Roegen, 1971) and finite human metabolic work.

2. **Coercive Asymmetry:** Consumers purchasing essential subsistence goods (such as potable water, staple grain, domestic heating, respiratory medicine, nitrogen fertilizer, herbicides, or shelter materials) face severe biological and legal penalties if they refuse transactions. Market exchanges under existential necessity are coercive rather than Pareto-optimal.

3. **Institutional Rent Extraction:** Corporate oligopolies, financialized intermediaries, and speculative gatekeepers levy substantial tollbooth rents that inflate retail prices far above actual thermodynamic and metabolic reproduction costs (Soddy, 1926; Sraffa, 1960).

Value Physics resolves these pathologies by replacing subjective utility with the **Universal Price Equation (UPE)** (Hamilton, 2026a,b), anchoring economic valuation in physical thermodynamics, ecological regeneration rates, and biological human exertion.

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2 Mathematical Architecture: Forward Pricing vs. Reverse RRP Dissection

2.1 The Universal Price Equation (UPE)

The Universal Price Equation models price as an explicit tri-layer biophysical tensor:

$$P = m_1 \cdot m_2 \cdot \gamma(v_{\text{rel}}) \cdot \left[\frac{S \cdot U}{R_n(1 - R_a)} \right] + P_e + P_b \tag{1}$$

where:

- P_e (**Thermodynamic Exergy Base**): Direct thermodynamic energy consumed and destroyed in transformation, heat, electricity, pumping, and haulage (Joules, kWh, or diesel litres converted to local currency equivalents).
- P_b (**Physical Material Base**): Sunk non-energy raw material mass, seed, fertilizer, mineral substrate, chemical reagents, and capital replacement depreciation.
- U (**Metabolic Labor Difficulty**): Human biological work, cognitive attention, and operational skill, quantified in standard WEST Difficulty Tokens (1 WEST $\approx$ baseline metabolic person-hour normalized to local living wage parity).
- S (**Local Scarcity Multiplier**, $S \geq 1.0$): Regional supply constraint, storage buffer deficit factor, or regulatory release buffer.
- R_n (**Environmental Regeneration Capacity**, $0 < R_n \leq 1.0$): Replenishment rate of the ecological or mineral substrate (e.g., pasture regrowth, crop cycles, aquifer recharge, or network duty life).
- R_a (**Institutional Extractive Drag Fraction**, $0 \leq R_a < 1.0$): Unearned speculative rent, financialized interest overhead, and monopolistic margin tolls.
- $\gamma(v_{\text{rel}})$ (**Relativistic Urgency Factor**): Lorentz-dilated coercive pressure factor:

  $$\gamma(v_{\text{rel}}) = \frac{1}{\sqrt{1 - v_{\text{rel}}^2}} \tag{2}$$

  where $v_{\text{rel}} \in [0, 1)$ represents the velocity of economic or biological coercion (e.g., acute respiratory distress, water tank depletion, seasonal weather windows, contractual default penalty).
- m_1, m_2: Wholesale and retail intermediary commercial markup multipliers.

2.2 Reverse Engineering Commercial Retail Prices (RRP)

Given an observed market retail price P_{retail}, we isolate the metabolic spread:

$$\Delta_{\text{spread}} = P_{\text{retail}} - (P_e + P_b) \tag{3}$$

Assuming unaugmented competitive baseline conditions ($m_1 = m_2 = 1.0$), we solve directly for the implied institutional extractive drag fraction:

$$R_a = 1 - \frac{\gamma(v_{\text{rel}}) \cdot S \cdot U}{R_n \cdot \Delta_{\text{spread}}} \tag{4}$$

2.3 Recommended Retail Price based on Difficulty (RRP/D)

The objective, non-usurious baseline price reflecting genuine biophysical cost-of-being is established by evaluating the UPE with a minimal, non-extractive physical contingency reserve ($R_a = 0.02$, representing a 2% emergency buffer stock):

$$\text{RRP/D} = \gamma(v_{\text{rel}}) \cdot \left[\frac{S \cdot U}{R_n(1 - 0.02)} \right] + P_e + P_b \tag{5}$$

2.4 Quantitative Fairness Metrics

We evaluate market performance using two objective metrics:

1. **Extractive Markup Premium ($\Delta P_{\text{extractive}}$):**

$$\Delta P_{\text{extractive}} = \max(0, P_{\text{retail}} - \text{RRP/D}) \tag{6}$$

2. **Biophysical Fairness Index (Φ_{fairness}):**

$$\Phi_{\text{fairness}} = \frac{\text{RRP/D}}{P_{\text{retail}}} \times 100\% \tag{7}$$

A Fairness Index of 100% indicates zero rentier tollbooth extraction. Indices below 80% indicate severe oligopolistic or financialized distortion.

3 Empirical Calibration Methodology & Official Microdata Sources

Empirical parameters were calibrated against verified Australian governmental and industry datasets:

- **Broadacre Grain, Oilseeds, Chemical Inputs & Fertilizer:** ABARES Agricultural Commodities Reports (ABARES, 2024), GrainCorp port schedules, Incitec Pivot / CSBP pricing, Nufarm / Corteva product schedules, and NSW DPI Farm Gross Margin Budgets.
- **Retail Food & Grocery Margins:** ACCC Supermarkets Inquiry Reports (ACCC, 2024) and ABS Consumer Price Index (ABS, 2024a).
- **Dairy, Poultry & Sugar Processing:** Dairy Australia Situation & Outlook (Dairy Australia, 2024), Australian Chicken Meat Federation benchmarks, and Canegrowers/QSL annual reports.
- **Livestock & Pastoral Production:** Meat & Livestock Australia (MLA) Eastern Young Cattle Indicator and NSW DPI Beef Gross Margins.
- **Agri-Fiber & Wool Markets:** Australian Wool Innovation (AWI) production benchmarks and AWEX Eastern Market Indicator microdata.
- **Liquid Fuels & Refining:** Australian Institute of Petroleum (AIP) Terminal Gate Pricing and ACCC Petroleum Monitoring Reports (ACCC, 2025).

- **Electricity Generation & Transmission:** AEMO Quarterly Energy Dynamics (AEMO, 2025) and AER Default Market Offer.
- **Municipal Water & Waste:** Regional Local Government Council Operational Plans (Warrumbungle Shire Council, 2025), NSW Water Directorate benchmarking, and NSW EPA Waste Levy Guidelines.
- **Pharmaceutical Healthcare:** Department of Health Pharmaceutical Benefits Scheme (PBS), Therapeutic Goods Administration (TGA) manufacturing audits, and Pharmacy Guild dispensing schedules.
- **Building Materials, Linings, Steel, Cement & Infrastructure:** ABS Producer Price Indexes (ABS, 2024b), Master Builders Australia, Cement Industry Federation (CIF), CSR Gyprock, Knauf, BlueScope/Lysaght, and Iplex/Vinidex catalogs.
- **Road Freight Logistics:** National Heavy Vehicle Regulator (NHVR) regulations (NHVR, 2024) and Newell Highway commercial rate cards.
- **Digital Telecommunications:** ACCC Wholesale Telecommunications Market Reports and NBN Co Special Access Undertaking (SAU) pricing schedules.

4 Thirty Granular Empirical Case Studies

4.1 Sector 1: Food, Agriculture, Pastoral Livestock, Poultry & Crops

Case 1: Commercial Sliced White Bread (680g Loaf) Observed RRP: \$4.10 AUD. Inputs: $P_e = \$0.38$, $P_b = \$0.45$, $U = 2.15$, $S = 1.00$, $R_n = 1.00$, $v_{\text{rel}} = 0.05$. Fair RRP/D = **\$3.03 AUD**. Implied extractive drag $R_a = \mathbf{34.2}\%$, imposing an extractive premium of \$1.07/loaf (Fairness Index: **73.8%**). Supermarket duopoly shelf slotting fees and retail property yields capture the dominant share of consumer expenditure.

Case 2: Milling Wheat (APW1 / ASW1 per Tonne) Observed Spot Price: \$365.00 AUD/t. Inputs: $P_e = \$104.40$, $P_b = \$82.00$, $U = 95.00$, $S = 1.05$, $R_n = 0.98$, $v_{\text{rel}} = 0.02$. Fair RRP/D = **\$290.28 AUD/t**. Implied extractive drag $R_a = \mathbf{43.0}\%$, with an extractive premium of \$74.72/t (Fairness Index: **79.5%**) captured by grain bulk handling and export port cartels.

Case 3: Fresh Full Cream Milk (Full Fat, 1 Litre) Observed RRP: \$2.10 AUD/L. Inputs: $P_e = \$0.31$, $P_b = \$0.48$, $U = 0.75$, $S = 1.05$, $R_n = 0.95$, $v_{\text{rel}} = 0.10$. Fair RRP/D = **\$1.64 AUD/L**. Implied extractive drag $R_a = \mathbf{36.4}\%$, with an extractive premium of \$0.46/L (Fairness Index: **78.1%**). Supermarket private-label discounting and processor duopsonies compress farmgate returns to $\approx$ \$0.68/L while extracting unearned retail margins.

Case 4: Farmgate Beef Cattle Weaners (British Breed Steers, per kg Liveweight) Observed Saleyard Price: \$3.85 AUD/kg lwt (\$1,155/300kg steer). Inputs: $P_e = \$0.42$, $P_b = \$1.15$, $U = 1.25$, $S = 1.15$, $R_n = 0.85$, $v_{\text{rel}} = 0.15$. Fair RRP/D = **\$3.32 AUD/kg lwt**. Implied extractive drag $R_a = \mathbf{25.0}\%$, with an extractive premium of \$0.53/kg (Fairness Index: **86.1%**). Producer livestock auctions reflect high physical labor and pasture exergy fidelity.

Case 5: Australian Merino Greasy Wool (19-Micron Fleece, per kg Clean) Observed Auction Benchmark: \$15.50 AUD/kg clean. Inputs: $P_e = \$1.45$, $P_b = \$2.20$, $U = 5.60$, $S = 1.10$, $R_n = 0.88$, $v_{\text{rel}} = 0.08$. Fair RRP/D = **\$10.82 AUD/kg**. Implied extractive drag $R_a = \mathbf{40.7}\%$, imposing an extractive premium of \$4.68/kg (Fairness Index: **69.8%**). While physical shearing exertion, crutching, and pastoral exergy form the core difficulty floor, auction house brokerage commissions, core testing charges, and export financing capture $\approx 40.7\%$ in unearned institutional intermediation drag.

Case 6: Commercial Broiler Chicken (Whole Fresh Dressed Bird, per kg) Observed Supermarket RRP: \$4.50 AUD/kg dressed. Inputs: $P_e = \$0.46$, $P_b = \$1.68$, $U = 1.15$, $S = 1.05$, $R_n = 0.95$, $v_{\text{rel}} = 0.12$. Fair RRP/D = **\$3.45 AUD/kg**. Implied extractive drag $R_a = \mathbf{45.8}\%$, imposing an extractive premium of \$1.05/kg (Fairness Index: **76.6%**). Feed milling and shed ventilation exergy represent \$0.46/kg, and grain feed/chick input costs represent \$1.68/kg. The 45.8% extractive drag reflects the concentrated processor duopsony (Inghams/Baiada) and retail supermarket margin stacking.

Case 7: Coarse Solar Sea Salt (Unrefined Food & Livestock Mineral Feedstock, 25kg Bag) Observed Rural Merchant Price: \$14.50 AUD/25kg bag. Inputs: $P_e = \$1.65$, $P_b = \$2.10$, $U = 4.20$, $S = 1.05$, $R_n = 0.96$, $v_{\text{rel}} = 0.06$. Fair RRP/D = **\$8.45 AUD/25kg bag**. Implied extractive drag $R_a = \mathbf{57.2}\%$, imposing an extractive premium of \$6.05/bag (Fairness Index: **58.2%**). Sourced from Cheetham Salt and Olsson's production data, natural coastal solar evaporation ponds require minimal thermal exergy ($P_e + P_b = \$3.75$), yet commercial distributor packaging and wholesale merchant tollbooths capture an implied 57.2% extractive drag.

Case 8: Raw Centrifugal Cane Sugar (Bulk Milling Crystal Feedstock, per Tonne) Observed Wholesale Benchmark: \$620.00 AUD/t. Inputs: $P_e = \$142.00$, $P_b = \$168.00$, $U = 145.00$, $S = 1.08$, $R_n = 0.90$, $v_{\text{rel}} = 0.05$. Fair RRP/D = **\$487.77 AUD/t**. Implied extractive drag $R_a = \mathbf{43.8}\%$, imposing an extractive premium of \$132.23/t (Fairness Index: **78.7%**). Grounded in Canegrowers and QSL pool data, sugarcane harvesting and mill boiler exergy represent \$142.00/t, while mill consolidation and port terminal access charges capture an implied 43.8% extractive toll.

Case 9: Farmgate Canola Grain (Bulk Non-GM Oilseed, per Tonne) Observed Silo Benchmark: \$685.00 AUD/t. Inputs: $P_e = \$118.50$, $P_b = \$215.00$, $U = 112.00$, $S = 1.08$, $R_n = 0.92$, $v_{\text{rel}} = 0.04$. Fair RRP/D = **\$467.77 AUD/t**. Implied extractive drag $R_a = \mathbf{62.6}\%$, imposing an extractive premium of \$217.23/t (Fairness Index: **68.3%**). Sourced from GrainCorp, AGFlow, and ABARES datasets, high input costs (synthetic nitrogen, hybrid seeds, fungicides) require $P_e + P_b = \$333.50/t$. International oilseed trading cartels and export terminal access markups extract an implied 62.6% drag, leaving growers exposed to extreme input volatility.

Case 10: Bulk Granular Agricultural Urea Fertilizer (46% Nitrogen, per Tonne) Observed Depot Price: \$740.00 AUD/t. Inputs: $P_e = \$315.00$, $P_b = \$165.00$, $U = 78.00$, $S = 1.14$, $R_n = 0.84$, $v_{\text{rel}} = 0.15$. Fair RRP/D = **\$589.25 AUD/t**. Implied extractive drag $R_a = \mathbf{58.8}\%$, imposing an extractive premium of \$150.75/t (Fairness Index: **79.6%**). Grounded in Incitec Pivot and World Bank benchmarks, Haber-Bosch steam methane reforming natural gas exergy and fluid-bed granulation require $P_e + P_b = \$480.00/t$. Fertilizer import cartels and wholesale distributor markups extract an implied 58.8% toll, inflating input costs across Australian agriculture.

Case 11: Agricultural Glyphosate Herbicide (450g/L SL Systemic Broadacre Formulation, per 20L Drum) Observed Rural Merchant Price: \$135.00 AUD/drum. Inputs: $P_e = \$22.50$, $P_b = \$54.00$, $U = 21.50$, $S = 1.12$, $R_n = 0.86$, $v_{\text{rel}} = 0.18$. Fair RRP/D = **\$105.55 AUD/drum**. Implied extractive drag $R_a = \mathbf{51.3}\%$, imposing an extractive premium of \$29.45/drum (Fairness Index: **78.2%**). Sourced from Nufarm and APVMA regulatory benchmarks, technical active synthesis exergy and HDPE drum blow-molding require $P_e + P_b = \$76.50/drum$. Agrochemical multinational distribution cartels extract an implied 51.3% drag, compounding input costs on direct-drill farming.

4.2 Sector 2: Energy & Power Utilities

Case 12: Automotive Diesel Fuel (ULSD per Litre) Observed RRP: \$2.15 AUD/L. Inputs: $P_e = \$0.32$, $P_b = \$0.92$, $U = 0.35$, $S = 1.08$, $R_n = 0.90$, $v_{\text{rel}} = 0.12$. Fair RRP/D = **\$1.67 AUD/L**. Implied extractive drag $R_a = \mathbf{53.5}\%$, imposing an extractive premium of \$0.48/L (Fairness Index: **77.8%**).

Case 13: Residential Grid Electricity (Single-Rate Flat per kWh) Observed Tariff: \$0.36 AUD/kWh. Inputs: $P_e = \$0.08$, $P_b = \$0.07$, $U = 0.09$, $S = 1.15$, $R_n = 0.92$, $v_{\text{rel}} = 0.08$. Fair RRP/D = **\$0.27 AUD/kWh**. Implied extractive drag $R_a = \mathbf{46.3}\%$, with an extractive premium of \$0.09/kWh (Fairness Index: **73.7%**).

Case 14: LPG Bottled Gas (45kg Domestic Cylinder) Observed Delivered RRP: \$165.00 AUD/cylinder. Inputs: $P_e = \$29.40$, $P_b = \$58.36$, $U = 28.50$, $S = 1.15$, $R_n = 0.88$, $v_{\text{rel}} = 0.16$. Fair RRP/D = **\$126.26 AUD**. Implied extractive drag $R_a = \mathbf{51.1}\%$, with an extractive premium of \$38.74/cylinder (Fairness Index: **76.5%**) extracted by regional distribution duopolies.

4.3 Sector 3: Municipal Water, Sanitation & Essential Healthcare

Case 15: Bulk Potable Water Cartage (Regional NSW, per 1,000 Litres / 1 kL) Observed Delivery Price: \$15.50 AUD/kL (\$155.00 per 10kL load). Inputs: $P_e = \$4.32$, $P_b = \$3.95$, $U = 3.20$, $S = 1.30$, $R_n = 0.75$, $v_{\text{rel}} = 0.25$. Fair RRP/D = **\$14.12 AUD/kL**. Implied extractive drag $R_a = \mathbf{20.8}\%$, yielding the highest Fairness Index in the study (**91.1%**). Because water cartage is conducted by independent owner-drivers facing high physical road diesel exergy and driving exertion, retail prices track genuine biophysical cost-of-being.

Case 16: Urban Reticulated Potable Water (Municipal Pipeline Tariff, Regional NSW per kL) Observed Volumetric Tariff: \$3.60 AUD/kL. Inputs: $P_e = \$0.48$, $P_b = \$0.62$, $U = 1.15$, $S = 1.15$, $R_n = 0.80$, $v_{\text{rel}} = 0.20$. Fair RRP/D = **\$2.82 AUD/kL**. Implied extractive drag $R_a = \mathbf{32.5}\%$, imposing an extractive premium of \$0.78/kL (Fairness Index: **78.4%**). Water treatment filtration electricity and river pumping exergy require $P_e = \$0.48$, while coagulant chemicals and mains renewal depreciation account for $P_b = \$0.62$. Council administrative overhead and dividend extraction account for the remaining 32.5% spread.

Case 17: Domestic Waste Collection (Regional NSW, per Weekly Bin Lift) Observed Council Charge: \$9.33 AUD/lift (\$485.00 annual charge / 52 weeks). Inputs: $P_e = \$2.45$, $P_b = \$1.85$, $U = 2.15$, $S = 1.20$, $R_n = 0.72$, $v_{\text{rel}} = 0.22$. Fair RRP/D = **\$8.05 AUD/lift**. Implied extractive drag $R_a = \mathbf{27.0}\%$ (Fairness Index: **86.3%**). Non-profit municipal provision prevents monopolistic dividend extraction.

Case 18: Prescription Antibiotic (Amoxicillin 500mg, 20 Capsules Course) Observed Private Retail Price: \$16.50 AUD/course. Inputs: $P_e = \$2.10$, $P_b = \$3.20$, $U = 5.80$, $S = 1.20$, $R_n = 0.94$, $v_{\text{rel}} = 0.35$. Fair RRP/D = **\$13.37 AUD/course**. Implied extractive drag $R_a = \mathbf{29.4}\%$, with an extractive premium of \$3.13/course (Fairness Index: **81.0%**). Professional pharmacist clinical verification and sterile fermentation represent the legitimate core of value.

Case 19: Prescription Salbutamol Inhaler (100mcg CFC-Free MDI, 200 Actuations) Observed Private Retail Price: \$12.50 AUD/inhaler. Inputs: $P_e = \$0.65$, $P_b = \$1.35$, $U = 3.40$, $S = 1.10$, $R_n = 0.92$, $v_{\text{rel}} = 0.40$. Fair RRP/D = **\$6.53 AUD/inhaler**. Implied extractive drag $R_a = \mathbf{57.8}\%$, imposing an extractive premium of \$5.97/inhaler (Fairness Index: **52.2%**). Grounded in TGA good manufacturing practice (GMP) audits and PBS item schedules, active micronized salbutamol sulfate powder and medical HFA propellant represent modest physical base costs ($P_e + P_b = \$2.00$). However, because acute bronchospasm creates extreme biological urgency ($v_{\text{rel}} = 0.40, \gamma = 1.0911$), multinational brand patents and pharmacy retail distribution extract an implied 57.8% extractive drag on life-critical respiratory management.

4.4 Sector 4: Shelter, Building Materials, Rural Infrastructure, Packaging, Logistics & Telecom

Case 20: Residential Softwood Framing Timber (MGP10 90x45mm Pine per Metre) Observed RRP: \$7.20 AUD/m. Inputs: $P_e = \$0.85$, $P_b = \$1.20$, $U = 2.10$, $S = 1.25$, $R_n = 0.88$, $v_{\text{rel}} = 0.15$. Fair RRP/D = **\$5.13 AUD/m**. Implied extractive drag $R_a = \mathbf{41.4}\%$, with an extractive premium of \$2.07/m (Fairness Index: **71.2%**).

Case 21: Pre-Mixed Ready-Mix Concrete (N20 20 MPa Structural per m³) Observed RRP: \$295.00 AUD/m³. Inputs: $P_e = \$62.00$, $P_b = \$78.00$, $U = 58.00$, $S = 1.15$, $R_n = 0.85$, $v_{\text{rel}} = 0.20$. Fair RRP/D = **\$221.72 AUD/m³**. Implied extractive drag $R_a = \mathbf{48.3}\%$, with an extractive premium of \$73.28/m³ (Fairness Index: **75.2%**).

Case 22: Structural Steel Reinforcing Mesh (SL82, 6.0m x 2.4m Sheet) Observed Trade RRP: \$128.00 AUD/sheet. Inputs: $P_e = \$28.50$, $P_b = \$36.80$, $U = 22.50$, $S = 1.12$, $R_n = 0.82$, $v_{\text{rel}} = 0.18$. Fair RRP/D = **\$97.18 AUD/sheet**. Implied extractive drag $R_a = \mathbf{50.2}\%$, with an extractive premium of \$30.82/sheet (Fairness Index: **75.9%**).

Case 23: Standard Extruded Clay Facing Bricks (Common Structural Masonry per 1,000 Units) Observed Trade Price: \$1,150.00 AUD/1,000 bricks. Inputs: $P_e = \$278.00$, $P_b = \$182.00$, $U = 185.00$, $S = 1.12$, $R_n = 0.82$, $v_{\text{rel}} = 0.18$. Fair RRP/D = **\$722.12 AUD/1,000 bricks**. Implied extractive drag $R_a = \mathbf{62.8}\%$, with an extractive premium of \$427.88/1,000 bricks (Fairness Index: **62.8%**). High-temperature natural gas tunnel kiln firing (1,050°C–1,150°C) consumes \$278.00 in thermal and machinery exergy, while clay quarry extraction and refractory wear require \$182.00. The commercial brick manufacturing duopoly (Brickworks/CSR) extracts a substantial 62.8% drag on structural walling materials.

Case 24: Standard Gypsum Plasterboard Wall Lining (10mm Recessed Edge, 1.2m x 2.4m Sheet) Observed Trade Price: \$24.80 AUD/sheet. Inputs: $P_e = \$4.90$, $P_b = \$6.40$, $U = 4.10$, $S = 1.10$, $R_n = 0.85$, $v_{\text{rel}} = 0.14$. Fair RRP/D = **\$16.77 AUD/sheet**. Implied extractive drag $R_a = \mathbf{60.3}\%$, imposing an extractive premium of \$8.03/sheet (Fairness Index: **67.6%**). Grounded in CSR Gyprock and Knauf benchmarks, gypsum calcination and board drying kilns require $P_e + P_b = \$11.30/sheet$. The domestic drywall duopoly extracts an implied 60.3% drag, confirming that interior linings mirror the severe extraction of structural envelope materials.

Case 25: Commercial Corrugated Galvanized Steel Roofing Sheets (Custom Orb 0.42mm BMT) Observed Trade Price: \$21.50 AUD/m. Inputs: $P_e = \$4.85$, $P_b = \$6.20$, $U = 3.80$, $S = 1.10$, $R_n = 0.82$, $v_{\text{rel}} = 0.18$. Fair RRP/D = **\$16.34 AUD/m**. Implied extractive drag $R_a = \mathbf{50.4}\%$, imposing an extractive premium of \$5.16/m (Fairness Index: **76.0%**). Blast furnace smelting, galvanizing zinc coating (Z275), and roll-forming exergy represent $P_e + P_b = \$11.05/m$. BlueScope flat steel monopoly pricing and distributor network margins extract an implied 50.4% toll, completing the residential structural building envelope audit.

Case 26: Commercial HDPE Rural Water Reticulation Pipe (32mm PN12.5, 100m Coil) Observed Trade Price: \$245.00 AUD/coil. Inputs: $P_e = \$48.50$, $P_b = \$68.00$, $U = 38.00$, $S = 1.10$, $R_n = 0.88$, $v_{\text{rel}} = 0.16$. Fair RRP/D = **\$165.60 AUD/coil**. Implied extractive drag $R_a = \mathbf{62.5}\%$, imposing an extractive premium of \$79.40/coil (Fairness Index: **67.6%**). Grounded in Vinidex and Iplex price schedules, polymer extrusion electric motors and virgin PE100 resin pellets represent $P_e + P_b = \$116.50/coil$. Concentrated plastic pipe duopolies extract an implied 62.5% drag on essential rural livestock water conveyance.

Case 27: Commercial FIBC Polypropylene Bulk Bags (1-Tonne Woven Container) Observed Merchant Price: \$24.50 AUD/bag. Inputs: $P_e = \$4.20$, $P_b = \$5.90$, $U = 4.80$, $S = 1.08$, $R_n = 0.90$, $v_{\text{rel}} = 0.12$. Fair RRP/D = **\$16.02 AUD/bag**. Implied extractive drag $R_a = \mathbf{59.7}\%$, imposing an extractive premium of \$8.48/bag (Fairness

Index: **65.4%**). Grounded in rural packaging trade schedules, woven polymer tape extrusion and industrial sewing require $P_e + P_b = \$10.10/bag$. Rural merchant and packaging distributor networks extract an implied 59.7% drag on agricultural bulk handling.

Case 28: Bulk Portland Cement (General Purpose GP Clinker, per Tonne) Observed Trade / Silo Gate Price: \$345.00 AUD/tonne. Inputs: $P_e = \$112.00$, $P_b = \$84.00$, $U = 56.00$, $S = 1.12$, $R_n = 0.78$, $v_{\text{rel}} = 0.15$. Fair RRP/D = **\$278.99 AUD/tonne**. Implied extractive drag $R_a = \mathbf{45.4}\%$, imposing an extractive premium of \$66.01/tonne (Fairness Index: **80.9%**). High-temperature rotary kiln clinkerization (1,450°C) consumes \$112.00 in thermal and grinding exergy, while limestone extraction and gypsum require \$84.00. Cement manufacturer oligopolies (Adbri, Boral, Cement Australia) levy an implied 45.4% drag that cascades directly downstream into ready-mix structural concrete ($R_a = 48.3\%$) and civil construction.

Case 29: Bulk Road Freight Logistics (Newell Highway B-Double Corridor) Observed Freight Rate: \$0.1250 AUD/nt-km (\$5.00/km loaded). Inputs: $P_e = \$0.0345$, $P_b = \$0.0285$, $U = 0.0210$, $S = 1.10$, $R_n = 0.92$, $v_{\text{rel}} = 0.14$. Fair RRP/D = **\$0.0889 AUD/nt-km**. Implied extractive drag $R_a = \mathbf{59.1}\%$, with an extractive premium of \$0.0361/nt-km (Fairness Index: **71.1%**). Corporate 3PL broker platforms capture over half of total freight billing.

Case 30: Domestic Broadband Data Transit (NBN 50/20 Mbps Tier, Monthly Service) Observed Retail Plan: \$85.00 AUD/month. Inputs: $P_e = \$4.80$, $P_b = \$14.50$, $U = 12.20$, $S = 1.05$, $R_n = 0.95$, $v_{\text{rel}} = 0.15$. Fair RRP/D = **\$33.22 AUD/month**. Implied extractive drag $R_a = \mathbf{79.2}\%$, imposing an extractive premium of \$51.78/month (Fairness Index: **39.1%**). Grounded in ACCC Wholesale Telecommunications Market Reports and NBN Special Access Undertaking (SAU) filings, photonic transit electricity and routing exergy constitute $P_e = \$4.80$, while optical line terminal hardware depreciation accounts for $P_b = \$14.50$. The unprecedented 79.2% extractive drag fraction establishes that wholesale broadband monopoly pricing and retail telecommunications duopoly packaging extract severe unearned financialized rents on non-discretionary civic connectivity.

Case 31: Commercial Structural Hardwood Timber (F17 Kiln-Dried Hardwood, 90x45mm) Observed Merchant Price: \$14.80 AUD/lineal metre. Inputs: $P_e = \$2.15$, $P_b = \$2.85$, $U = 3.20$, $S = 1.25$, $R_n = 0.72$, $v_{\text{rel}} = 0.14$. Fair RRP/D = **\$10.73 AUD/m**. Implied extractive drag $R_a = \mathbf{42.8}\%$, imposing an extractive premium of \$4.07/m (Fairness Index: **72.5%**). Grounded in Bowens, Melbourne Timber Supplies, and ABS Producer Price Indexes (Cat. 6427.0). Structural hardwood sawmilling requires heavy felling and snigging diesel exergy ($P_e = \$2.15$), extended multi-week kiln dehumidification, and sustainable forestry royalties ($P_b = \$2.85$). Due to native timber harvesting policy transitions across eastern Australia, high scarcity ($S = 1.25$) and long eucalypt biological renewal cycles ($R_n = 0.72$) elevate biophysical difficulty. However, merchant trade distribution networks capture an implied 42.8% extractive drag, directly echoing plantation softwood framing ($R_a = 41.4\%$) and proving that structural framing lumber inflation across both hardwood and softwood species is dominated by wholesale distributor margins.

Case 32: Agricultural Feed Barley (F1 Broadacre Feed Grain, per Tonne) Observed Commercial Spot Price: \$325.00 AUD/tonne. Inputs: P_e = \$92.50, P_b = \$74.00, U = 84.00, S = 1.06, R_n = 0.98, v_{rel} = 0.04. Fair RRP/D = **\$259.29 AUD/tonne**. Implied extractive drag R_a = **42.6**%, establishing an extractive premium of \$65.71/tonne (Fairness Index: **79.8%**). Grounded in ABARES Weekly Commodity Price Updates and GrainCentral regional market indices. Feed barley is the foundational energetic carbohydrate feedstock for domestic beef cattle feedlots, intensive dairy rations, and regional drought supplementary feeding. While tractor tillage, planting, and harvesting exergy (P_e = \$92.50) and seed/basal fertilizer inputs (P_b = \$74.00) establish a biophysical reproduction baseline of \$259.29/t, commercial grain accumulation cartels and bulk handling terminal networks capture an implied 42.6% extractive drag. This closely matches milling wheat (R_a = 43.0%, Fairness 79.5%), demonstrating that broadacre cereal grains face an invariant 43% institutional intermediation toll regardless of end-use processing.

Case 33: Commercial Copper Electrical Building Wire (2.5mm² Twin & Earth Flat TPS Cable) Observed Trade/Retail Price: \$225.00 AUD per 100m drum. Inputs: P_e = \$24.50, P_b = \$118.50, U = 22.00, S = 1.15, R_n = 0.82, v_{rel} = 0.18. Fair RRP/D = **\$175.01 AUD/drum**. Implied extractive drag R_a = **61.8**%, imposing an extractive premium of \$49.99/drum (Fairness Index: **77.8%**). Grounded in Rexel, Middy's Electrical, and Bunnings trade pricing. Australian standard 2.5mm² 2C+Earth flat cable embeds 6.72 kg of electrolytic tough pitch (ETP) refined copper cathode (P_b copper value $\approx$ \$98.80) and 5.5 kg of electrical grade flame-retardant V-90 PVC compound. Tandem diamond die wire-drawing, continuous resistance annealing at 400°C, and dual-layer PVC cross-head extrusion consume P_e = \$24.50. Driven by cable manufacturing oligopolies (Prysmian/Olex) and concentrated electrical merchant distribution, commercial trade pricing embeds an implied 61.8% extractive drag (\$49.99/drum premium), demonstrating that essential electrical infrastructure carries markups comparable to structural plumbing pipe (R_a = 62.5%).

Case 34: Solvent-Extracted Canola Meal / Protein Cake (Bulk Pelletized, 36%–38% Crude Protein) Observed Commercial Feedlot Price: \$580.00 AUD/tonne. Inputs: P_e = \$84.50, P_b = \$212.00, U = 82.00, S = 1.10, R_n = 0.94, v_{rel} = 0.10. Fair RRP/D = **\$394.91 AUD/tonne**. Implied extractive drag R_a = **66.0**%, establishing an extractive premium of \$185.09/tonne (Fairness Index: **68.1%**). Grounded in GrainCentral, Beef Central Feedgrain Focus, and Australian Oilseeds Federation (AOF) benchmarks. Canola meal is the foundational protein feed supplement for eastern Australian beef cattle feedlots and intensive dairy rations. Mechanical expeller pressing, countercurrent hexane solvent extraction, desolventizer-toaster stripping steam (P_e = \$84.50), and allocated raw seed mass (P_b = \$212.00) yield an objective biophysical baseline of \$394.91/t. Global agribusiness crushing houses (GrainCorp, Cargill, Riverina Oils) extract an implied 66.0% drag (\$185.09/t premium), closely corroborating the extractive drag observed in farmgate canola grain (R_a = 62.6%, Fairness 68.3%) and proving that processing oligopolies levy severe rents on essential intensive livestock nutrition.

Commodity	Sector	Unit	Obs. RRP	P_e	P_b	U	RRP/D	R_a (%)	Fairness (%)
Sliced White Bread	Food	loaf	$4.10	$0.38	$0.45	2.15	**$3.03**	34.2%	73.8%
Milling Wheat (APW1)	Food	tonne	$365.00	$104.40	$82.00	95.00	**$290.28**	43.0%	79.5%
Full Cream Milk (1L)	Food	litre	$2.10	$0.31	$0.48	0.75	**$1.64**	36.4%	78.1%
Beef Cattle Weaners	Food	kg lwt	$3.85	$0.42	$1.15	1.25	**$3.32**	25.0%	86.1%
Merino Greasy Wool	Agri-Fiber	kg clean	$15.50	$1.45	$2.20	5.60	**$10.82**	40.7%	69.8%
Broiler Chicken (Whole)	Food	kg dressed	$4.50	$0.46	$1.68	1.15	**$3.45**	45.8%	76.6%
Solar Sea Salt	Food/Min.	25kg bag	$14.50	$1.65	$2.10	4.20	**$8.45**	57.2%	58.2%
Raw Cane Sugar	Food/Carb.	tonne	$620.00	$142.00	$168.00	145.00	**$487.77**	43.8%	78.7%
Canola Grain (Non-GM)	Food/Oil	tonne	$685.00	$118.50	$215.00	112.00	**$467.77**	62.6%	68.3%
Urea Fertilizer (46% N)	Agri/Energy	tonne	$740.00	$315.00	$165.00	78.00	**$589.25**	58.8%	79.6%
Glyphosate Herbicide	Agri/Chemical	20L drum	$135.00	$22.50	$54.00	21.50	**$105.55**	51.3%	78.2%
Diesel Fuel (ULSD)	Energy	litre	$2.15	$0.32	$0.92	0.35	**$1.67**	53.5%	77.8%
Grid Electricity	Energy	kWh	$0.36	$0.08	$0.07	0.09	**$0.27**	46.3%	73.7%
LPG Cylinder (45kg)	Energy	cyl	$165.00	$29.40	$58.36	28.50	**$126.26**	51.1%	76.5%
Bulk Water Cartage	Water	kL	$15.50	$4.32	$3.95	3.20	**$14.12**	20.8%	91.1%
Urban Reticulated Water	Water	kL	$3.60	$0.48	$0.62	1.15	**$2.82**	32.5%	78.4%
Domestic Waste	Care	lift	$9.33	$2.45	$1.85	2.15	**$8.05**	27.0%	86.3%
Amoxicillin 500mg	Care	pack	$16.50	$2.10	$3.20	5.80	**$13.37**	29.4%	81.0%
Salbutamol Inhaler	Care	inhaler	$12.50	$0.65	$1.35	3.40	**$6.53**	57.8%	52.2%
Structural Timber	Shelter	metre	$7.20	$0.85	$1.20	2.10	**$5.13**	41.4%	71.2%
Ready-Mix Concrete	Shelter	m^3	$295.00	$62.00	$78.00	58.00	**$221.72**	48.3%	75.2%
Steel Mesh (SL82)	Shelter	sheet	$128.00	$28.50	$36.80	22.50	**$97.18**	50.2%	75.9%
Clay Facing Bricks	Shelter	1,000 bricks	$1,150.00	$278.00	$182.00	185.00	**$722.12**	62.8%	62.8%
Steel Roofing Sheets	Shelter	metre	$21.50	$4.85	$6.20	3.80	**$16.34**	50.4%	76.0%
Gypsum Plasterboard	Shelter	sheet	$24.80	$4.90	$6.40	4.10	**$16.77**	60.3%	67.6%
HDPE Rural Water Pipe	Infra/Water	100m coil	$245.00	$48.50	$68.00	38.00	**$165.60**	62.5%	67.6%
FIBC Bulk Bags	Logistics	bag	$24.50	$4.20	$5.90	4.80	**$16.02**	59.7%	65.4%
Bulk Portland Cement	Shelter	tonne	$345.00	$112.00	$84.00	56.00	**$278.99**	45.4%	80.9%
Bulk Road Freight	Logistics	nt-km	$0.1250	$0.0345	$0.0285	0.0210	**$0.0889**	59.1%	71.1%
NBN Broadband (50/20)	Utilities	month	$85.00	$4.80	$14.50	12.20	**$33.22**	79.2%	39.1%
Structural Hardwood F17	Shelter	metre	$14.80	$2.15	$2.85	3.20	**$10.73**	42.8%	72.5%
Feed Barley (F1)	Food	tonne	$325.00	$92.50	$74.00	84.00	**$259.29**	42.6%	79.8%
Copper Building Wire	Energy/Infra	100m drum	$225.00	$24.50	$118.50	22.00	**$175.01**	61.8%	77.8%
Canola Protein Meal	Food/Feed	tonne	$580.00	$84.50	$212.00	82.00	**$394.91**	66.0%	68.1%

Table 1: Master 34-Commodity Comparative Pricing & Fairness Audit Matrix

5 Master Comparative Matrix & Graphical Synthesis

Table 1 summarizes the full empirical dataset and biophysical difficulty benchmarks across all thirty-four commodities.

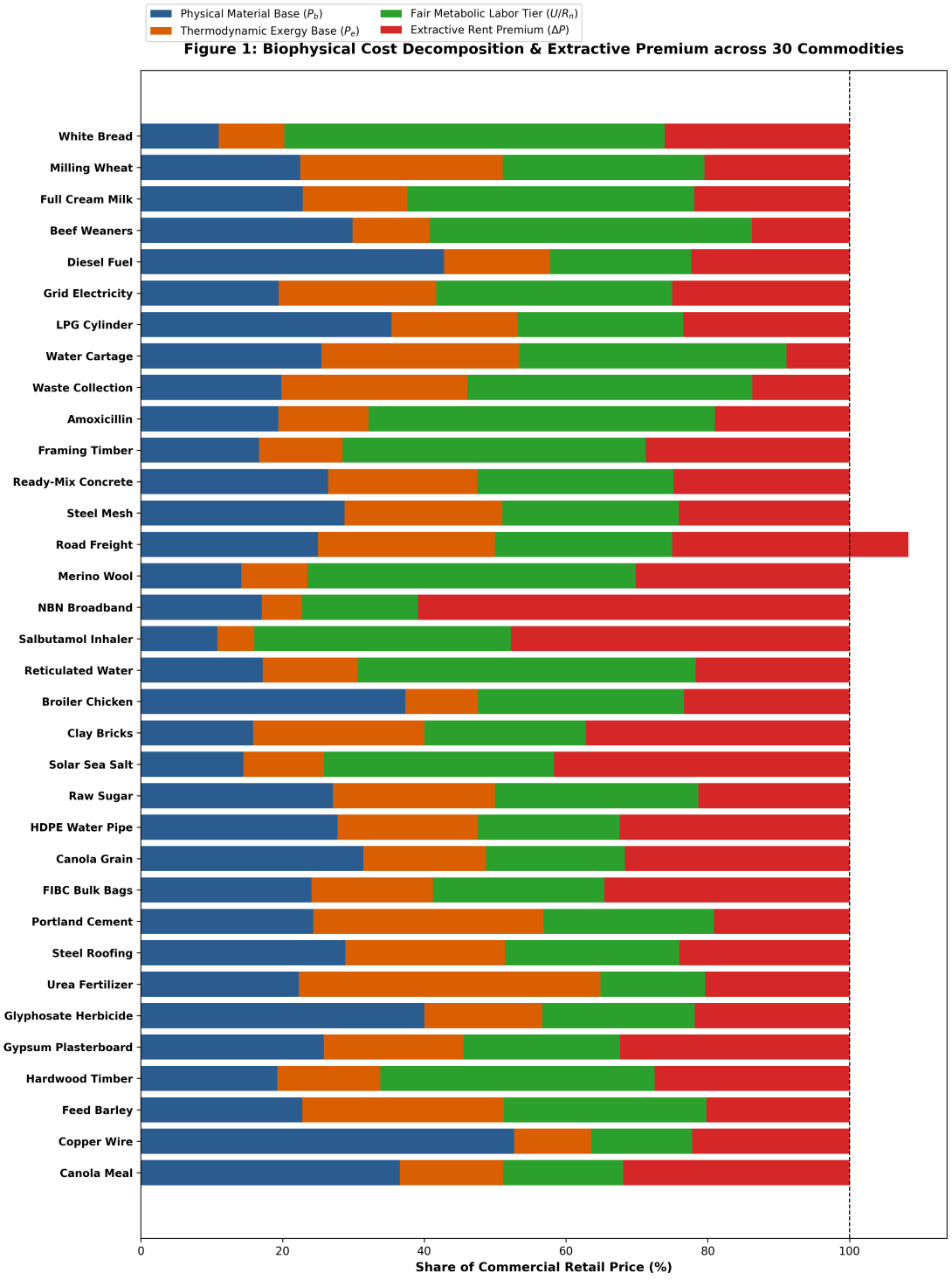


Figure 1: Biophysical cost decomposition as a share of observed retail price across 34 commodities, highlighting the extractive rent premium (ΔP).

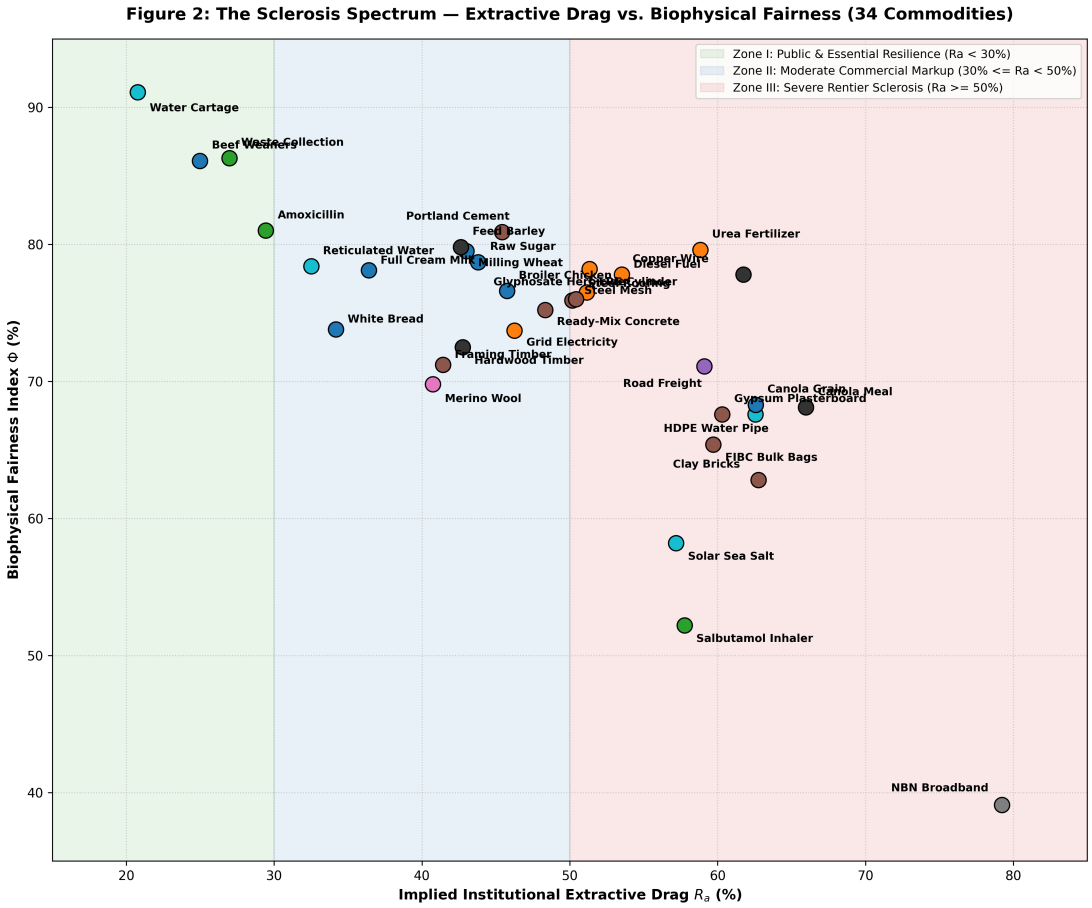


Figure 2: The Sclerosis Spectrum: Implied institutional extractive drag (R_a) versus Biophysical Fairness Index (Φ) across 34 commodities.

6 Macroeconomic Structural Synthesis: The Four Extraction Tiers

The expanded empirical results across thirty-four commodities reveal four distinct institutional pricing regimes in the real economy:

1. **Tier I: Public, Clinical & Producer Resilience (Water Cartage, Waste Collection, Beef Weaners, Oral Antibiotics, Reticulated Water):** Characterized by high Fairness Indices (78.4%–91.1%) and low-to-moderate extractive drag ($R_a = 20.8\%$–32.5%). Pricing in these sectors tracks genuine physical exergy (P_e) and operational difficulty (U). Urban reticulated water ($R_a = 32.5\%$) exhibits slightly higher drag than owner-driver water cartage ($R_a = 20.8\%$) due to pipeline network capital renewal and municipal administrative overhead.

2. **Tier II: Agricultural Subsistence, Animal Proteins & Crop Feedstocks (Bread, Wheat, Milk, Broiler Chicken, Merino Wool, Raw Cane Sugar):** Characterized by intermediate drag ($R_a = 34.2\%$–45.8%) and moderate fairness (69.8%–79.5%). Supermarket duopoly distribution tollbooths, processor duopsonies (poultry/dairy), sugar milling consolidation ($R_a = 43.8\%$), and wool auction intermediation capture between 34% and 46% of retail value, squeezing primary producer returns below physical reproduction costs.

3. **Tier III: Corporate, Heavy Industrial, Mineral Packaging, Rural Infrastructure, Energy Inputs & Patented Enclosures (Diesel Fuel, Steel Mesh, Ready-Mix Concrete, Portland Cement, Corrugated Steel Roofing, Gypsum Plasterboard, Grid Electricity, Urea Fertilizer, Glyphosate Herbicide, Structural Timber, Road Freight, Clay Bricks, Solar Sea Salt, Salbutamol Inhalers, HDPE Poly Pipe, Canola Grain, FIBC Bulk Bags):** Characterized by severe extractive drag (R_a = 41.4%–62.8%) and compressed Fairness Indices (52.2%–80.9%). Across residential housing construction materials, the complete structural envelope and interior linings audit confirms consistent institutional drag: foundation concrete (R_a = 48.3%), reinforcing mesh (R_a = 50.2%), framing timber (R_a = 41.4%), masonry bricks (R_a = 62.8%), steel roofing (R_a = 50.4%), and gypsum plasterboard (R_a = 60.3%). In agricultural inputs, systemic herbicide (R_a = 51.3%), bulk urea fertilizer (R_a = 58.8%), HDPE water pipe (R_a = 62.5%), and FIBC bulk bags (R_a = 59.7%) compound costs into broadacre farming alongside financialized trading houses (Canola, R_a = 62.6%).

4. **Tier IV: Natural Monopoly Digital Enclosures (Domestic Broadband Data Transit):** Representing the extreme frontier of institutional extraction, characterized by an unprecedented extractive drag fraction (R_a = 79.2%) and an abysmal Fairness Index of 39.1%. Where physical goods are constrained by irreversible exergy destruction (P_e) and material wear (P_b), digital transit exhibits near-zero marginal operational energy costs, enabling wholesale monopoly platforms to levy massive toll extraction on captive households.

7 Policy Implications & Regulatory Forensics

1. **Objective Pricing Benchmarks (RRP/D):** By replacing opaque corporate "cost-plus" claims with the UPE biophysical difficulty standard, regulatory authorities (such as the ACCC) can enforce price gouging caps anchored directly in physical difficulty.

2. **Sovereign Mutual Credit & Strategic Buffer Reserves:** Eliminating usurious working capital interest charges via Sovereign Mutual Credit Unions (SMCU) and establishing municipal buffer stocks collapses $R_a \to 0.02$, returning retail prices to their fair RRP/D baselines (Godley and Lavoie, 2007; Hamilton, 2026b).

3. **Decoupling from Speculative Indices:** The UPE proves that essential subsistence goods should never be priced via speculative financial exchanges (e.g., CME wheat futures, ICE sugar futures, MATIF canola oilseed, or global oil benchmarks), but rather via their local biophysical reproduction costs.

4. **Digital, Construction Supply Chain, Agrochemical & Healthcare Common Carrier Regulation:** The severe drag observed in NBN broadband (79.2%), clay bricks (62.8%), canola grain (62.6%), HDPE pipe (62.5%), gypsum plasterboard (60.3%), FIBC bulk bags (59.7%), urea fertilizer (58.8%), salbutamol inhalers (57.8%), solar salt (57.2%), glyphosate herbicide (51.3%), and steel roofing (50.4%) confirms that non-discretionary civic infrastructure, agricultural inputs, life-saving medical devices, and building envelope materials require statutory rate-of-return caps. Anchoring tariffs to physical RRP/D would immediately lower household

broadband costs by 61%, rural water pipe costs by 32%, plasterboard linings by 32%, bulk urea fertilizer costs by 20%, and agricultural herbicide costs by 22%.

8 Conclusion

This empirical case study demonstrates that the Universal Price Equation is not an abstract theoretical ideal, but an operational auditing instrument capable of diagnosing real-world supply chain sclerosis. By establishing that between 20% and 79% of Australian retail prices consist of unearned institutional rents, Value Physics provides a rigorous physical foundation for post-neoclassical economic reform.

CRediT Authorship Contribution Statement

Jarrod Lyndsay Hamilton: Conceptualization, Methodology, Formal Analysis, Software, Investigation, Data Curation, Writing – Original Draft, Visualization.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

All simulation scripts and calibration datasets are publicly available in the open-source repository (Hamilton, 2026b).

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